ANEES HUSSAIN

EXCELLENCE IN EDUCATION SINCE 1989

XI PHYSICS

PHYSICS -H.S.C. PART-I (PRE-ENGINEERING & MEDICAL)

According to the new syllabus of the Intermediate Board of Education, Karachi

MULTIPLE CHOICE QUESTIONS

CHAPTER 1

PHYSICS AND MEASUREMENTS

1. **$[MLT^{-2}]$ is dimensional formula of**
   * Pressure * **Force** * Displacement * Strain
2. **Dimensions of kinetic energy is the same as that of:**
   * Force * Velocity * **Work** * Acceleration
3. **The respective number of significant figures for the numbers 23.023, 0.0003 and 2.1×10^{-3} are:**
   * 4, 4,2 * 5,5,2 * 5, 1, 5 * **5, 1, 2**
4. **Systematic error can be:**
   * Zero error * **Either positive or negative**
   * Positive only * Negative only
5. **The unit of solid angle is**
   * Candela * Kilogram * **Steradian** * Second
6. **What is the SI unit of electric current?**
   * Watts * Ohms * **Amperes** * Volts
7. **What is the speed of light in a vacuum approximately?**
   * 50 mph * 2.5×10^{17} km/h * **3×10^{8} m/s** * 9.8 m/s^2
8. **Which instrument is used to measure electric voltage?**
   * Ohmmeter * Oscilloscope * **Ammeter** * Voltmeter
9. **Which of the following is a fundamental force of nature?**
   * Tension force * **Gravitational force**
   * Magnetic force * Frictional force
10. **What is the SI unit of energy?**
    * Pascal * **Joule** * Watt * Newton

S.NO#	M.C.Q'S	S.NO#	M.C.Q'S
1	Force	6	Amperes
2	Work	7	3×10^8 m/s
3	5, 1, 2	8	Ammeter
4	Either positive or negative	9	Gravitational force
5	Steradian	10	Joule

KINEMATICS

1. **The velocity of a particle at an instant is 10 m/s and after 5sec the velocity of particle is 20m/s. The velocity 3 sec before in m/s is:**
   * 7 * 6 * **4** * 8
2. **A ball is thrown upwards with a velocity of 100 m/s. It will reach the ground after**
   * 40 sec * 5 sec * **20 sec** * 10 sec
3. **At what angle the range of projectile becomes equal to the height of projectile?**
   * 30º * 45º * **76º** * 65º
4. **The ratio of numerical values of average velocity and average speed of a body is always:**
   * Less than unity * **Unity or less** * Unity * None of these
5. **If the dot product of two non-zero vectors vanishes, the vectors will be:**
   * zero * **perpendicular to each other**
   * opposite direction to each other * in the same direction
6. **What is the equation that relates velocity (v), initial velocity (u), acceleration (a), and time (t) in one dimension?**
   * v = u – at * v = u / at * v = u x at * **v = u + at**
7. **If an object moves with a constant velocity, what can you say about its acceleration?**
   * Acceleration is decreasing. * Acceleration is increasing.
   * Acceleration is constant * **Acceleration is zero**.
8. **A car accelerates from rest at a rate of 2 m/s². How long does it take to reach a velocity of 10 m/s?**
   * 10 seconds * **5 seconds** * 4 seconds * 2 seconds
9. **An object is dropped from a height. What is the acceleration due to gravity on Earth?**
   * 6m/s² * 10 m/s² * **9.8 m/s²** * 8.2 m/s²
10. **What is the formula to calculate displacement (s) in terms of initial velocity (u), final velocity (v), and time (t)?**
    * s = (u / v)t * s = (u x v)t * s = (u - v)t * **s = (u + v)t**

S.NO#	M.C.Q'S	S.NO#	M.C.Q'S
1	4	6	v = u + at
2	20 sec	7	Acceleration is zero
3	76º	8	5 seconds
4	Unity or less	9	9.8 m/s²
5	perpendicular to each other	10	s = (u + v)t

DYNAMICS

1. **The term mass refers to the same physical concept as**
   * Acceleration * Force * **Inertia** * Weight
2. **The laws of motion show the relationship between**
   * **Force and acceleration** * Mass and acceleration
   * Mass and velocity * Velocity and acceleration
3. **The kinetic energy of body of mass 2kg and momentum of 2 Ns is:**
   * 4 J * 3 J * 2 J * **1 J**
4. **A 3kg bowling ball experiences a net force of 15N. What will be its acceleration.**
   * 35 m/s^2 * **5 m/s^2** * 7 m/s^2 * 35 m/s^2
5. **The rate of change of linear momentum of a body is called a:**
   * **Impulse** * Power * Angular force * Linear force
6. **What is the fundamental concept behind Newton's second law of motion?**
   * Action and reaction are equal and opposite * Conservation of energy
   * **Force equals mass times acceleration** * Conservation of momentum
7. **A car is traveling at a constant speed on a straight road. What can you conclude about the net force acting on the car?**
   * The net force is changing
   * The net force is opposite to the direction of motion
   * The net force is in the direction of motion
   * **The net force is zero**
8. **Which of the following is a unit of force in the International System of Units (SI)?**
   * Pascal * Watt * **Newton** * Joule
9. **When an object is in equilibrium, which of the following statements is true?**
   * The object has no acceleration * The object is at rest
   * **The net force on the object is zero** * The object has no velocity
10. **According to Newton's third law of motion, for every action, there is an equal andopposite reaction. What type of pair is this?**
    * Inertia forces * **Action-reaction forces**
    * Balanced forces * Non-interacting forces

S.NO#	M.C.Q'S	S.NO#	M.C.Q'S
1	Inertia	6	Force equals mass times acceleration
2	Force and acceleration	7	The net force is zero
3	1 J	8	Newton
4	5 m/s^2	9	The net force on the object is zero
5	Impulse	10	Action-reaction forces

ROTATIONAL AND CIRCULAR MOTION

1. **One radian is about**
   * **57º** *37 º * 45 º * 25 º
2. **The rotational inertia of a wheel about its axle does not depend upon its**
   * **speed of rotation** * mass * distribution of mass * diameter
3. **An object rotating about a fixed axis, I is its rotational inertia and α is its angular acceleration. Its**
   * is the definition of torque * is the definition of rotational inertia
   * is definition of angular acceleration * **follows directly from Newton's second law**
4. **A stone of 2 kg is tied to a 0.50 m long string and swung around a circle at angular velocity of 12 rad/s. The net torque on the stone about the center of the circle is**
   * **0 Nm** * 12 Nm * 6Nm * 72 Nm
5. **If the external torque acting on a body is zero, then its**
   * Angular momentum is zero * **Angular momentum is conserved**
   * Angular acceleration is maximum * Rotational motion is maximum
6. **Which of the following quantities is a vector in rotational motion?**
   * **Angular velocity** * Angular displacement
   * Angular acceleration * Moment of inertia
7. **When an object undergoes uniform circular motion, what remains constant?**
   * **Speed** * Velocity * Acceleration * Angular displacement
8. **What is the SI unit of angular velocity?**
   * m/s *m/s^2 ***rad/s** * Hz
9. **If the net torque acting on an object is zero, what can be said about its angular velocity?**
   * **It is constant** * It is increasing * It is decreasing * It is zero
10. **The moment of inertia of an object depends on which of the following factors?**
    * **Mass and shape** * Mass and velocity
    * Shape and density * Velocity and density

S.NO#	M.C.Q'S	S.NO#	M.C.Q'S
1	57º	6	Angular velocity
2	Speed of rotation	7	Speed
3	Follows directly from Newton's second law	8	rad/s
4	0 Nm	9	It is constant
5	Angular momentum is conserved	10	Mass and shape

WORK, ENERGY AND POWER

1. **The kinetic energy of a light and a heavy object is the same. Which object has maximum momentum?**
   * Light object * **Heavy object**
   * Both have same momentum * N.O.T
2. **A body fallen from height h. After it has fallen a height h/2, its will possess:**
   * only potential energy * only kinetic energy
   * **half potential half kinetic energy** * more kinetic and less
3. **When one joule of work is done on a body in one second, power of body is said to**
   * **One watt** * 0.5 watt * zero * 100 watt
4. **The absolute potential energy of an object depends on:**
   * The object's mass and height * **The object's mass and speed**
   * The object's shape and size * The object's color and
5. **The escape velocity of a planet depends on which of the following factors?**
   * The mass of the planet only * The radius of the planet only
   * **Both the mass and the radius of the planet** * The density of the planet
6. **Which of the following is a scalar quantity?**
   * **Work** * Energy * Power * Force
7. **The unit of energy is:**
   * Newton (N) * **Joule (J)**
   * Watt (W) * Meter per second (m/s)
8. **If an object is lifted vertically upwards, which type of work is done on the object?**
   * **Positive work** * Negative work * Zero work * None
9. **Which law of thermodynamics states that energy cannot be created or destroyed, only converted from one form to another?**
   * **First law of thermodynamics** * Second law of thermodynamics
   * Third law of thermodynamics * Zeroth law of thermodynamics
10. **Power is defined as:**
    * The rate of change of force * **The rate of doing work**
    * The rate of acceleration * The rate of change of energy

S.NO#	M.C.Q'S	S.NO#	M.C.Q'S
1	*Heavy object*	**6**	*Work*
2	*half potential half kinetic energy*	**7**	*Joule (J)*
3	*One watt*	**8**	*Positive work*
4	*The object's mass and speed*	**9**	*First law of thermodynamics*
5	*Both the mass and the radius of the planet*	**10**	*The rate of doing work*

FLUID STATICS

1. **The pressure exerted on the ground by a man is greatest when:**
   * He stands with both feet flat on the ground * **He stands on the toes of one foot**
   * He stands flat on one foot * He lies down on the ground
2. **One piston in a hydraulic lift has an area that is twice the area of the other. When the pressure at the smaller piston is increased by Δp the pressure at the larger piston:**
   * Increases by 2Δp * **Increases by Δp**
   * Increases by Δp/2 * Increases by 4Δp
3. **The pressure at the bottom of a pond does NOT depend on:**
   * Water density * **The surface area of the pond**
   * The depth of the pond * None of these
4. **"An object completely submerged in a fluid displaces its own volume of fluid". This is:**
   * Pascal's paradox * **Pascal's principle**
   * Archimedes' principle * True, but none of the above
5. **You fill a tall glass with ice and then add water to the level of the glass's rim, so some fraction of the ice floats above rim. When the ice melts, what's happens to thewater level?**
   * The water overflows the rim * The water level drops below the rim
   * **The water level stays at the rim**
   * It depends on the difference in density between water and ice
6. **What property of a fluid determines its buoyant force on an object submerged in it?**
   * **Density of the fluid** * Viscosity of the fluid
   * Pressure of the fluid * Temperature of the fluid
7. **When an object is completely submerged in a fluid and in equilibrium, what is trueabout the buoyant force and the weight of the object?**
   * Buoyant force is greater than the weight of the object
   * Buoyant force is less than the weight of the object
   * **Buoyant force is equal to the weight of the object**
   * Buoyant force depends on the shape of the object
8. **Archimedes' principle states that the buoyant force on an object is equal to:**
   * **The weight of the fluid displaced by the object**
   * The volume of the fluid displaced by the object
   * The density of the fluid
   * The pressure at the bottom of the object
9. **What is the SI unit of pressure?**
   * **Pascal (Pa)** * Newton (N) * Joule (J) * Kilogram (kg)
10. **In a fluid at rest, how does the pressure vary with depth?**
   * **Pressure increases with depth** * Pressure decreases with depth
   * Pressure remains constant with depth * Pressure depends on the density of fluid

S.NO#	M.C.Q'S	S.NO#	M.C.Q'S
1	He stands on the toes of one foot	6	Density of the fluid
2	Increases by Δp	7	Buoyant force is equal to the weight of the object
3	The surface area of the pond	8	The weight of the fluid displaced by the object
4	Pascal's principle	9	Pascal (Pa)
5	The water level stays at the rim	10	Pressure increases with depth

CHAPTER 07

FLUID DYNAMICS

1. **Bernoulli's principle states that for horizontal flow of a fluid through a tube, the sumof the pressure and energy of motion per unit volume is**
   * Increasing with time * **Constant**
   * Decreasing with time * Varying with time
2. **Which of the following is associated with the law of conservation of energy in fluids?**
   * Archimedes' principle * Pascal's principle
   * **Bernoulli's principle** * equation of continuity
3. **If the cross-sectional area of a pipe decreases, what happens to the fluid velocity?**
   * **Increases** * Remains the same
   * Decreases * Depends on the fluid
4. **Wind speeding up as it blows over the top of a hill**
   * Increases atmospheric pressure there
   * **Decreases atmospheric pressure there**
   * Doesn't affect atmospheric pressure there
   * Equal's atmospheric pressure.
5. **A fluid is undergoing steady flow. Therefore:**
   * The velocity of any given molecule of fluid does not change.
   * The pressure does not vary from point to point
   * **The velocity at any given point does not vary with time**
   * The density does not vary from point to point
6. **What is the property of a fluid that describes its resistance to flow?**
   * Density * **Viscosity** * Pressure * Surface tension

7. **According to Bernoulli's equation, which of the following quantities remains constantalong a streamline of fluid flow?**
   * **Pressure** * Velocity * Density * Volume
8. **When fluid flows through a pipe, which of the following factors can increase the flow rate?**
   * Increasing pipe length * **Increasing pipe diameter**
   * Increasing fluid viscosity * Decreasing fluid density
9. **What type of flow occurs when the fluid particles move in parallel and at a constant velocity, with no crossing of streamlines?**
   * Turbulent * **Laminar** * Compressible * Irrotational
10. **Which principle explains why a ship can float on water?**
   * Pascal's principle * **Archimedes' principle**
   * Boyle's law * Newton's third law

S.NO#	M.C.Q'S	S.NO#	M.C.Q'S
1	Constant	6	Viscosity
2	Bernoulli's principle	7	Pressure
3	Increases	8	Increasing pipe diameter
4	Decreases atmospheric pressure there	9	Laminar
5	The velocity at any given point does not vary with time	10	Archimedes' principle

ELECTRIC FIELDS

1. **Which of the following can be deflected while moving in the electric field?**
   * Neutron * Photon * Electron * **Electron and proton**
2. **The flux through a flat surface of area "A" in a uniform electric field "E" is maximumwhen the surface area is**
   * **Parallel to E** * Perpendicular to E* Placed 45° to E * Placed 60° to E
3. **The negative gradient of the potential is:**
   * Potential energy * Voltage * **Electric field intensity** * Electric flux
4. **The electric flux through a plane area will be half of its maximum value when area isheld at an angle of____with an electric field.**
   * 30° * **60°** * 45° * Electric flux
5. **A 2μC point charge is located a distance "d" away from 6μC point charge, what is the ratio of F_{12}/F_{21}?**
   * 1/3 * 3 * **1** * 12
6. **What is the SI unit of electric field intensity?**
   * Volts * Coulombs * **Newtons per meter** * Amperes
7. **When a positive charge is placed in an electric field, in which direction does it experience a force?**
   * Opposite to the direction of the field * **Parallel to the direction of the field**
   * Perpendicular to the direction of the field * None of the above
8. **What is the formula for the electric field intensity (E) due to a point charge Q at a distance r from it?**
   * **$E = kQ/r^2$** * $E = kQ/r$ * $E = Q/r^2$ * $E = Q/k^2$
9. **If you double the distance between a point charge and a test charge, how does the electric field strength at the new location compared to the original?**
   * It becomes four times weaker * **It becomes half as strong**
   * It remains the same * It becomes twice as strong
10. **In which direction does an electron move when placed in an electric field?**
   * **Opposite to the direction of the field** * Parallel to the direction of the field
   * Perpendicular to the direction of the field * None of the above

S.NO#	M.C.Q'S	S.NO#	M.C.Q'S
1	Electron and proton	6	Newtons per meter
2	Parallel to E	7	Parallel to the direction of the field
3	Electric field intensity	8	$E = kQ/r^2$
4	60°	9	It becomes half as strong
5	1	10	Opposite to the direction of the field

CAPACITANCE

1. **What is the value of capacitance of a capacitor which has a voltage of 4V and has 16C of charge?**
   * **2F** * 4F * 6F * 8F
2. **In a variable capacitor, capacitance can be varied by:**
   * **Turning the rotatable plates in or out** * Changing the plates
   * Sliding the rotatable plates * Changing the material of plates
3. **The charging of a capacitor through a resistance follows**
   * Linear law * Square law * **Exponential law** * None
4. **The capacitance C is charged through a resistor R. The time constant of the charging circuit is given by**
   * C/R * 1/RC * **RC** * R/C
5. **Capacitor blocks:**
   * Alternating current * Direct current
   * Both alternating and direct current * **Neither alternating nor direct current**
6. **What is the SI unit of capacitance?**
   * Ohm * **Farad** * Volt * Ampere
7. **The ability of a capacitor to store electric charge is determined by its**
   * Voltage * Resistance * **Capacitance** * Current
8. **In a parallel plate capacitor, if the distance between the plates is doubled, how does the capacitance change?**
   * It doubles * **It becomes half** * It quadruples * It remains the same
9. **When a dielectric material is inserted between the plates of a capacitor, what happens to the capacitance?**
   * It decreases * **It increases** * It remains the same * It becomes zero
10. **If the potential difference across a capacitor is 12 volts and it stores a charge of 48 microcoulombs, what is its capacitance?**
   * **4 F** * 12 F * 48 F * 144 F

S.NO#	M.C.Q'S	S.NO#	M.C.Q'S
1	*2F*	**6**	*Farad*
2	*Turning the rotatable plates in or out*	**7**	*Capacitance*
3	*Exponential law*	**8**	*It becomes half*
4	*RC*	**9**	*It increases*
5	*Neither alternating nor direct current*	**10**	*4 F*

D.C CIRCUITS

1. **The resistance of a superconductor is:**
   * Finite * Infinite * Changes with every conductor * **Zero**
2. **The graphical representation of Ohms law is:**
   * Parabola * Hyperbola * Ellipse * **Straight line**
3. **A potential difference is applied across the ends of a wire. If the potential difference is doubled, then the drift velocity of free electrons will**
   * Be quadrupled * **Be doubled** * Be halved * Remain unchanged
4. **Power dissipation in a resistor can be calculated using which formula:**
   * $P=V^2/R$ * $P=I^2R$ * $P=VxI$ * **all of these**
5. **A heat-sensitive device whose resistivity changes with the change in temperature is called:**
   * Conductor * Resistor * **Thermistor** * Thermometer
6. **What is the SI unit of electric current?**
   * Volt * Watt * **Ampere** * Ohm
7. **In a series circuit, how does the total resistance (R_E) change as you add more resistors?**
   * **R_E total increases** * R_E total decreases
   * R_E total remains the same * It depends on the values of the resistors
8. **Ohm's Law relates which three quantities in a DC circuit?**
   * **Voltage (V), Current (I), and Resistance (R)** * Voltage (V), Power (P), and Energy (E)
   * Current (I), Capacitance (C), and Inductance (L)
   * Voltage (V), Frequency (f), and Wavelength (λ)
9. **If a resistor has a resistance of 100 ohms and a current of 0.5 amperes flowing through it, what is the voltage across the resistor?**
   * **50 V** * 200 V * 0.005 V * 0.2 V
10. **Kirchhoff's first law, also known as the law of conservation of electric charge, states that:**
   * The total voltage in a closed loop is zero
   * The total current in a closed loop is zero
   * The total resistance in a closed loop is zero
   * **The total charge in a closed loop is constant**

S.NO#	M.C.Q'S	S.NO#	M.C.Q'S
1	*Zero*	**6**	*Ampere*
2	*Straight line*	**7**	*R_E total increases*
3	*Be doubled*	**8**	*Voltage (V), Current (I), and Resistance (R)*
4	*All of these*	**9**	*50 V*
5	*Thermistor*	**10**	*The total charge in a closed loop is constant*

OSCILLATIONS

1. **A spring attached by a load of weight W is vibrating with a period T. If the spring is divided in four equal parts and the same load is suspended from one of these parts, the new time period is:**
   * T/4 * 2T * **T/2** * 4T
2. **The total energy of a particle executing simple harmonic motion is proportional to**
   * Frequency of oscillation * Maximum velocity of motion
   * Amplitude of motion * **Square of amplitude of motion**
3. **If a body oscillates at the angular frequency ωd of the driving force, then the oscillations are called:**
   * **Forced oscillations** * Coupled oscillations * Free oscillations * Maintained oscillations
4. **In vehicles, shock absorber reduced the jerks:**
   * The shock absorber is the application of damped oscillations.
   * Damping effect is due to the fractional loss of energy
   * Shock absorbers in vehicles reduced jerk * **All of these**
5. **A heavily damped system has a fairly flat resonance curve in:**
   * An acceleration time graph * **An amplitude frequency graph**
   * Velocity time graph * Distance-time graph
6. **What is the SI unit of frequency?**
   * **Hertz (Hz)** * Newton (N) * Pascal (Pa) * Joule (J)
7. **Which of the following quantities describes the maximum displacement of anoscillating object from its equilibrium position?**
   * Frequency * **Amplitude** * Phase * Period
8. **A simple pendulum oscillates with a period T. If its length is doubled, what happens to its period?**
   * T remains the same * T becomes half * T becomes four times * **T becomes double**
9. **The time it takes for one complete cycle of oscillation is known as:**
   * Amplitude * Frequency * **Period** * Wavelength
10. **In simple harmonic motion (SHM), the restoring force is:**
   * Directly proportional to velocity * **Directly proportional to displacement**
   * Constant * Proportional to acceleration

S.NO#	M.C.Q'S	S.NO#	M.C.Q'S
1	T/2	6	Hertz (Hz)
2	Square of amplitude of motion	7	Amplitude
3	Forced oscillations	8	T becomes double
4	All of these	9	Period
5	An amplitude frequency graph	10	Directly proportional to displacement

ACOUSTICS

1. **Two sound waves are y = A sin(ωt - kx) and y = A cos(ωt - kx). The phase difference between the two waves is:**
   * **π/2** * π/4 * π * 0
2. **If v_a, v_h and v_m are the speeds of sound in air, hydrogen gas, and a metal at the same temperature, then:**
   * $v_a > v_h > v_m$ * **$v_m > v_h > v_a$**
   * $v_h > v_m > v_a$ * $v_h > v_a > v_m$
3. **The speed of sound in a gas is proportional to:**
   * Square root of isothermal elasticity * **Square root of adiabatic elasticity**
   * Isothermal elasticity * Adiabatic elasticity
4. **The speed of sound in a gas in which two waves of wavelength 50 cm and 50.4 cm produce 6 beats per second is:**
   * **303 m/s** * 350 m/s * 378 m/s * 400 m/s
5. **The speed of a wave in a medium is 760 m/s. If 3600 waves are passing through a point in the medium in 2 minutes, then its wavelength is**
   * 13.85 m * **25.3 m** * 41.5 m * 57.2 m
6. **What is the speed of sound in air at room temperature (approximately)?**
   * 300,000 m/s * 1,000 m/s * **343 m/s** * 3,000,000 m/s
7. **Which of the following characteristics of a sound wave determines its pitch?**
   * Amplitude * **Frequency** * Wavelength * Speed
8. **Sound waves are examples of which type of waves?**
   * Transverse waves * Electromagnetic waves
   * **Longitudinal waves** * Standing waves
9. **What is the unit of sound intensity?**
   * **Decibel (dB)** * Hertz (Hz) * Pascal (Pa) * Joule (J)
10. **When sound waves reflect off a surface, it is called:**
    * Refraction * Diffraction * Absorption * **Reflection**

S.NO#	M.C.Q'S	S.NO#	M.C.Q'S
1	π/2	6	343 m/s
2	$v_m > v_h > v_a$	7	Frequency
3	Square root of adiabatic elasticity	8	Longitudinal waves
4	303 m/s	9	Decibel (dB)
5	25.3 m	10	Reflection

PHYSICAL OPTICS

1. **If the wavelength of an electromagnetic wave is about the diameter of a cricket ball, what type of radiation is it.**
   * X-ray * Ultraviolet * **Radio waves** * Visible light
2. **In Young's experiment when the distance between slits and screen is doubled, while separation of slits is halved, then fringe width will be:**
   * **4 times** * ¼ times * Doubled * Unchanged
3. **A ray of light passes from air into water. Striking the surface of the water with an N.angle of incidence 45°. Which of these quantities change as the light enters the water. i) Wavelength ii) frequency iii) speed of propagation iv) direction of propagation**
   * i and ii only. * iii and iv only. * **i, iii, and iv only** * All of them.
4. **Optically active substances are those substances which**
   * Produce polarized light * **Rotate the plane of polarization of polarized light**
   * Produce double refraction * Convert a plane polarized light into circularly polarized light
5. **Plane polarized light is passed through a Polaroid. On viewing through the Polaroid, wefind that when Polaroid is given one complete rotation about the direction of light**
   * The intensity of light gradually decreases to zero and remains at zero.
   * The intensity of light gradually increases to maximum and remains at maximum
   * There is no change in the intensity of light
   * **The intensity of light varies such that it is twice maximum and twice zero**
6. **What term describes the bending of light waves as they pass from one medium into another with a different refractive index?**
   * Reflection * Dispersion * **Refraction** * Diffraction
7. **Which phenomenon explains why a rainbow displays a spectrum of colors?**
   * Diffraction * **Dispersion** * Interference * Reflection
8. **When a beam of light travels from a denser medium to a rarer medium, which of thefollowing is true?**
   * The speed of light decreases, and its frequency increases
   * **The speed of light increases, and its frequency decreases**
   * The speed of light decreases, and its wavelength decreases
   * The speed of light increases, and its wavelength increases
9. **What is the phenomenon where light waves cancel each other out, resulting in areas of darkness in the pattern of light?**
   * Polarization * **Interference** * Dispersion * Diffraction
10. **Which type of lens can converge parallel rays of light to a single focal point?**
   * **Convex lens** * Concave lens * Plano-concave lens * Plano-convex lens

S.NO#	M.C.Q'S	S.NO#	M.C.Q'S
1	**Radio waves**	**6**	**Refraction**
2	**4 times**	**7**	**Dispersion**
3	**i, iii, and iv only**	**8**	**The speed of light increases, and its frequency decreases**
4	**Rotate the plane of polarization of polarized light**	**9**	**Interference**
5	**The intensity of light varies such that it is twice maximum and twice zero**	**10**	**Convex lens**

CHAPTER 14

COMMUNICATION

1. A communication channel consists of:
   * Transmission line only * **Optical fiber only**
   * Free space only * All of them
2. As compared to sound waves frequency of radio waves is:
   * **Higher** * Equal * Lower * May be higher or lower
3. The process of superimposing signal frequency on the carrier wave is known as
   * Transmission * Detection * Reception * **Modulation**
4. AM is used for broadcasting because:
   * Requires less transmitting power compare with other systems
   * It is more noise immune than other modulation system
   * No other modulation can provide the necessary hand width faithful transmission
   * **Its use avoids receiver complexity**
5. Data in compact disc is stored in the form of
   * Analog signal * **Digital signal** * Noise * Both (a) &(b)
6. What type of electromagnetic waves is commonly used for wireless basedcommunication, including radio and television?
   * Gamma rays * Infrared waves * **Microwaves** * X-rays
7. Which property of a wave represents the number of oscillations (cycles) per unit timeand is used to specify the frequency of a signal?

* Amplitude * Wavelength * Phase * **Frequency**

8. In optical fiber communication, data is typically transmitted using:
   * Infrared light * Ultraviolet light * **Visible light** * X-rays
9. What does the term "modulation" refer to in the context of communication systems?
   * **The process of encoding information onto a carrier wave**
   * The process of decoding received signals
   * The process of amplifying signals
   * The process of filtering out noise from signals
10. Which of the following communication technologies relies on the reflection of radio waves to communicate over long distances?
   * Fiber optics * **Satellite communication**
   * Bluetooth * Infrared communication

S.NO#	M.C.Q'S	S.NO#	M.C.Q'S
1	**Optical fiber only**	**6**	**Microwaves**
2	**Higher**	**7**	**Frequency**
3	**Modulation**	**8**	**Visible light**
4	**Its use avoids receiver complexity**	**9**	**The process of encoding information onto a carrier wave**
5	**Digital signal**	**10**	**Satellite communication**